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**004604**

May 2025

**B. Tech. CSE (AI & ML) (Sixth Semester)**

**Software Engineering (PEC-CS-S-601)**

*Time : 3 Hours]* *[Maximum Marks : 75*

**Note :** It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

## Part A

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1. (a) Define software engineering and its importance. **1.5**

   (b) Explain the term 'Silver Bullet' in software development. **1.5**

   (c) What is the primary advantage of the Agile Model over the Waterfall Model ? **1.5**

   (d) What are the components of an SRS document ? **1.5**

   (e) Define Function Point (FP) estimation technique. **1.5**

(f) Describe Behavioural modelling and Data modelling 1.5

(g) Discuss risk analysis and management. 1.5

(h) Describe the concept of software verification and validation. 1.5

(i) Explain error, bug, defect, fault, failure. 1.5

(j) What is Software Quality Assurance (SQA) ? Explain its importance. 1.5

**Part B**

2. (a) Differentiate between Spiral and RAD Model. 5

(b) Differentiate between Personal Software Process (PSP), Team Software Process (TSP) and also explain the emergence of software engineering. 10

3. (a) What are the different estimation techniques used in software project management ? Discuss LOC-based and FP-based estimation. 10

(b) A project size of 200 KLOC is to be developed. Software development team has average experience of similar type of projects. The project schedule is not very

tight. Calculate the effort, development time, average staff size, productivity of the project. The basic COCOMO coefficients are given in the table : 5

| Project | a | b | c | d |
|---|---|---|---|---|
| Organic | 2.4 | 1.05 | 2.5 | 0.38 |
| Semi-detached | 3.0 | 1.12 | 2.5 | 0.35 |
| Embedded | 3.6 | 1.20 | 2.5 | 0.32 |

4. (a) Given a situation where a software project failed due to lack of proper requirement gathering, suggest improvements based on requirement engineering principles. 5

(b) Explain the four steps of the "Requirements Engineering Process". Differentiate between functional and non-functional requirements. 10

5. (a) Explain the concepts of coupling and cohesion with examples. 5

(b) State when a module can be called functionally independent of other modules. Why functional independence is the key factor for a good software design ? Differentiate between cohesion and coupling. Mention the different types of coupling and cohesion in modules. 10

6. (a) Consider the program to find the greatest number 10

```
#include<stdio.h>
main( )
    {
    Float x,y,z;
    Clrscr( ),
    Printf("Enter three varaibles x,y,z");
    Scanf("%f%f%f", &x,&y,&z);
    If(x>y)
    {
        If(x>z)
            Printf("x is greatest");
        Else
            Printf("z is greatest")
    }
    Else
    {
        If(y>z)
            Printf("y is greatest");
        Else
            Printf("z is greatest")
    }
Getch( )
}
```

(a) Draw the DD graph for the program

(b) Calculate the cyclomatic complexity of program.

(c) List all independent paths.

(d) Design all the test cases from independent paths.

(b) Differentiate between Unit testing, Integration testing, Acceptance Testing. 5

7. Write short notes on the following : 15

(a) CMM

(b) The ISO 9126 Standard

(c) Formal approaches to SQA

MAY 2023

B.Tech (ENC) 6th SEMESTER

Software Engineering (PEC-CS-S-601)

Time: 3 Hours

Max. Marks:75

*Instructions:*
1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*
2. *Answer any four questions from Part -B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

## PART –A

Q1 (a) Discuss 'No Silver Bullet' in Software Engineering. (1.5)

(b) Differentiate between Personal Software Process and Team Software Process. (1.5)

(c) Explain Modularity. What problems are likely to arise if two modules have high coupling? (1.5)

(d) Explain Risk management process? (1.5)

(e) Differentiate beween LOC and FP based estimation technique. (1.5)

(f) What is Behavioral Modeling? Explain (1.5)

(g) Differentiate between Verification and Validation. (1.5)

(h) How Alpha Testing is different from Beta Testing. (1.5)

(i) Explain Statistical Software Quality Assurance. (1.5)

(j) Discuss ISO 9126 standard. (1.5)

## PART –B

Q2 (a) What do you understand by Agile Software Development Life Cycle?? Discuss the problems in Waterfall Model and the need of Agile Software Development model. (10)

(b) Explain Software process, project and product. (5)

Q3 (a) Draw the Level-0, Level-1 and Level-2 DFD for Travel Agency Management System. Clearly describe the working of the system. Also mention all assumptions made by you. (10)

(b) Discuss the process of Requirement Engineering. Also explain the different technique Of Software Elicitation. (5)

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Q4 What are the different categories of software development projects according to the COCOMO estimation model? Give examples of software development projects belonging to each of these categories. (15)

Q5 (a) Discuss the meaning of term modular design. What are the different factors that affect the modularity of a design? How can you assess the modularity of a design? (10)

(b) Explain the characteristics of a good software design. (5)

Q6 (a) Differentiate between Static Testing and Dynamic Testing. (5)

(b) Consider a program to determine largest of three numbers A, B and C within the range [1, 50] and prints the largest number. Design boundary value test cases by all the three techniques of Boundary Value Analysis. (10)

Q7 (a) List five salient requirements that a software development organization must comply with before it can be awarded with ISO 9126 certificate. (10)

(b) Illustrate SQA Activities. Define the purpose of Quality Assurance in software development. (5)